Dynamical Ising machines can solve discrete tomography problem

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Dynamical Ising machines are continuous dynamical systems that evolve from a generic initial state to a state strongly related to the ground state of the classical Ising model. We show that an Ising machine, driven by the V_2 dynamical model, can exactly solve discrete tomography problems about reconstructing a binary image from the pixel sums along a discrete set of rays. For the problems with at most two rays intersecting at each pixel, the randomly initialized V_2 model converges in polynomial time to a state with high probability representing an image precisely satisfying the tomographic data. Such a convergence is provided by the model's ability to undergo transitions of the discrete component of the relaxed spins beyond Hamming-neighborhood constraints. Our consideration demonstrates that Ising machines driven by specific dynamical systems can produce exact solutions to highly non-trivial data processing tasks.

I. INTRODUCTION

Ising machines represent an approach to computing that utilizes inherent properties of networks of classical spins, taking values $\sigma = \pm 1$. These computational properties attract researchers' attention for a long time [1–4], and stem from two key properties of the Ising model ground state. First, the ground state can be regarded as solving a quadratic unconstrained binary optimization problem [5]. Second, by associating the spin distribution in the ground state with a partition of the graph nodes, one obtains the maximum (weighted) cut of the model graph [6]. Finding the maximum cut is an NP-complete problem [7, 8], which puts the Ising model into the general computing perspective [9].

In dynamical Ising machines [10], the computational properties of classical spin networks are exposed through representing the networks in various continuous dynamical systems designed or selected with the objective that the system progresses towards a state that is tightly related to the network ground state or, equivalently, to the maximum cut partitioning of the network graph. This is achieved by representing the binary spins by continuous variables, $\sigma_m \to \xi_m$, and associating with the collective state of the dynamical variables a Lyapunov function, constrained by two requirements. First, it should change monotonously on the trajectories of the dynamical system. Second, spin-like states, with $\boldsymbol{\xi} = \boldsymbol{\sigma}$, yield the same value of the objective function as its discrete counterpart. For example, if the dynamical system is characterized by a continuous energy function, $\mathcal{H}_M(\boldsymbol{\xi})$, then $\mathcal{H}_M(\boldsymbol{\xi} = \boldsymbol{\sigma}) = \mathcal{H}(\boldsymbol{\sigma})$, where $\mathcal{H}(\boldsymbol{\sigma})$ is the Ising energy.

Requiring that the objective function monotonously changes with evolution puts Ising machines into the context of solving optimization problems, which, at present, is a dominating approach to Ising machines. However, the optimization perspective is too broad, and the discrete optimization is famous for ubiquitous hard problems [11, 12]. For example, the general max-cut problem, which is native to Ising machines, besides being NP-hard is also APX-hard [13], meaning that there is a limit to approximation that can be achieved in polynomial time on a sufficiently rich family of problems. Consequently, the Ising machines are commonly regarded as heuristic optimization engines aiming for approximate solutions, which makes it challenging to put Ising machines into the context of solving "common" information processing tasks that demand exact solutions, even if non-deterministic.

In the present paper, we demonstrate that individual dynamical models driving Ising machines possess a powerful computational resource that allows them to solve certain non-trivial data-processing problems exactly. Specifically, we show that an Ising machine driven by the V_2 dynamical model [14] starting from a randomized initial state evolves towards a state that represents a solution to the discrete tomography problem about restoring a binary image from its tomographic data, the collection of pixel sums along a discrete set of rays. Our consideration shows that Ising machines can be regarded in a much broader context as platforms for realizations of an alternative class of viable algorithms performing "traditional" data-processing tasks.

It must be emphasized in this regard that the sole convergence of any Ising machine to a state representing a solution of a data-processing task is an expected consequence of the machine's operating principles and a proper representation of the task as finding a ground state of an Ising model. However, such principles do not warrant that the machine will demonstrate practical scaling properties with increasing the problem size. For example, as we demonstrate below, an attempt to solve the discrete tomography problem using local search may result in almost-exponential decay of the solving probability. It is, therefore, important that the polynomial scaling exhibited by the V_2 machine is the consequence of the specific dynamical features of the V_2 model, in particular, the ability to traverse spin configurations transcending Hamming-bound constraints.

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The rest of the paper is organized as follows. In Section II, we overview the main properties of the V_2 model. In Section III, we provide the graph-theoretical formulation of the discrete tomography problem to solve by the V_2 model. In Section IV, we analyze the convergence of the V_2 model to the solutions of the tomography problem. Finally, in Section V, we show the results of numerical simulations.

II. BASIC PROPERTIES OF THE V_2 MODEL

The V_2 model describes a special class of dynamical systems with dissipative dynamics. The dynamical variables of the system are associated with nodes $\mathscr{V}$ of undirected finite graph $\mathscr{G} = \{\mathscr{V}, \mathscr{E}\}$, whose set of edges $\mathscr{E}$ with assigned weights (both, positive and negative weights are allowed) represent coupling between the variables. We will denote the distribution of variables associated with the graph nodes by $\boldsymbol{\xi} = (\xi_1, \ldots, \xi_N)$. The evolution of the system is governed by equations of motion that are convenient to define as $\dot{\boldsymbol{\xi}} = 2\nabla_{\boldsymbol{\xi}} \mathcal{C}_{\mathrm{V}_2}(\boldsymbol{\xi})$, ensuring that $\mathcal{C}_{\mathrm{V}_2}(\boldsymbol{\xi})$ is a non-decreasing function of time. The V_2 model is defined by

$$\mathcal{C}_{\mathrm{V}_2}(\boldsymbol{\xi}) = \frac{1}{4} \sum_{\alpha,\beta \in \mathscr{V}} A_{\alpha,\beta} |\xi_\alpha - \xi_\beta|_{[-2,2]}, \tag{1}$$

where $A_{\alpha,\beta}$ is the (weighted) adjacency matrix of graph $\mathscr{G}$, and $|\ldots|_{[-2,2]}$ is a periodic function with period 4 defined for $-2 \leq \xi \leq 2$ by $|\xi|_{[-2,2]} = |\xi|$. Taking $\xi_\alpha = \pm 1$ turns $\mathcal{C}_{\mathrm{V}_2}(\boldsymbol{\xi})$ into the total weight of edges between the nodes corresponding to ξ's with different signs. In other words, $\mathcal{C}_{\mathrm{V}_2}(\boldsymbol{\xi})$ is a relaxation of the cut function.

The connection with the combinatorial structures emerging on the graph and the V_2 model is made more transparent by employing the periodicity of the kernel in Eq. (1) and introducing new dynamical variables, relaxed spins, by the relation $\xi_\alpha = \sigma_\alpha + X_\alpha + 4k_\alpha$. Here, $\sigma \in \{-1, 1\}$ and $X \in (-1, 1]$ are the binary and continuous components of the relaxed spin, respectively. Integer k_α accounts for the kernel periodicity in $\mathcal{C}_{\mathrm{V}_2}(\boldsymbol{\xi})$ and plays no role in the following. Rewriting Eq. (1) using the relaxed spin representation, we obtain

$$\mathcal{C}_{\mathrm{V}_2}(\boldsymbol{\sigma}, \mathbf{X}) = \mathcal{C}(\boldsymbol{\sigma}) + \widetilde{\mathcal{C}}_{\mathrm{V}_2}(\boldsymbol{\sigma}, \mathbf{X}), \tag{2}$$

with

$$\widetilde{\mathcal{C}}_{\mathrm{V}_2}(\boldsymbol{\sigma}, \mathbf{X}) = \frac{1}{4} \sum_{\alpha,\beta} A_{\alpha,\beta} \sigma_\alpha \sigma_\beta |X_\alpha - X_\beta|. \tag{3}$$

Respectively, the dynamics of relaxed spins is governed for $X_\alpha \in [-1, 1)$ by

$$\dot{X}_\alpha = \sum_{\beta \in \mathscr{V}} A_{\alpha,\beta} \sigma_\alpha \sigma_\beta \operatorname{sgn}(X_\alpha - X_\beta), \tag{4}$$

where $\operatorname{sgn}(X)$ is the sign function with the adopted convention $\operatorname{sgn}(0) = 0$. The evolution described by these equations occurs on the phase space of relaxed spins illustrated by Fig. 1. The topology of the phase space is reflected by supplementing the equations of motion with the rule of negotiating the basepoint in the wedge sum by the phase point: whenever X_α crosses the $X = \pm 1$ boundary, σ_α changes sign.

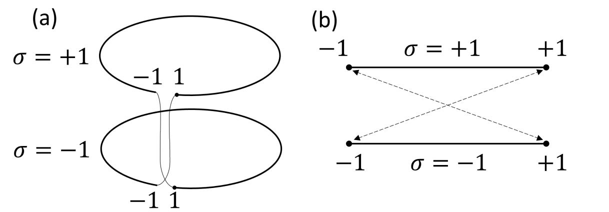

FIG. 1. The phase space of relaxed spins in the V_2 model as (a) two circles with the transition rule, or, alternatively, as (b) two intervals $[-1, 1]$ with respectively identified boundary points.

The computational significance of the evolution of dissipative dynamical Ising machines can be established by relating it to the properties of the terminal states of the machine initiated in a generic state. The dynamics governed by Eqs. (4) terminates at critical points of $\mathcal{C}_{\mathrm{V}_2}(\boldsymbol{\sigma}, \mathbf{X})$ that possess several general key features discussed in [14]).

First of all, while the terminal states of the V_2 model are not necessarily binary (the continuous components of the relaxed spins may not vanish in the terminal state), it is trivial to round them optimally because of the vanishing term $\widetilde{\mathcal{C}}_{\mathrm{V}_2}(\boldsymbol{\sigma}, \mathbf{X})$ at critical points and the invariance of the relaxed cut function $\widetilde{\mathcal{C}}_{\mathrm{V}_2}(\boldsymbol{\sigma}, \mathbf{X})$ with respect to global homogeneous translation $\mathbf{X} \to \mathbf{X} + r\mathbf{1}$, with $\mathbf{1} = (1, 1, \ldots, 1)$ on the model phase space. Hence, nonbinary solutions can be mapped to binary states by simply ignoring the continuous component of the relaxed spin. It is worth noting that this is a specific property of the V_2 model. Various relaxations can be productively represented in terms of relaxed spins with the relaxed cut function invariant with respect to global phase rotations. However, the binary states produced by such rotations may yield different cut values (because of non-vanishing $\widetilde{\mathcal{C}}_{\mathrm{V}_2}(\boldsymbol{\sigma}, \mathbf{X})$) raising the special problem of optimal rounding (see, e.g., Section 8.2 in [15]).

The terminal states of the V_2 model have a characteristic structure with relaxed spins grouping into clusters with the same value of the continuous component. This is a consequence of the fact that for states with all continuous component different, the relaxed cut function is a linear function of $\mathbf{X}$ with the vanishing Hessian. To account for this circumstance, we introduce the notion of *strong clusters.* Critical point $(\boldsymbol{\sigma}, \mathbf{X})$ partitions the set of graph nodes $\mathscr{V}$ into a system of $1 \leq S \leq |\mathscr{V}|$ disjoint subsets (strong clusters) $\mathscr{V} = \bigcup_{p=1}^{S} \mathscr{V}^{(p)}(\boldsymbol{\sigma}, \mathbf{X})$ with $\mathscr{V}^{(p)}(\boldsymbol{\sigma}, \mathbf{X}) = \left\{\alpha \in \mathscr{V} : X_\alpha = X^{(p)}\right\}$ and $X^{(p)} \neq X^{(p')}$ for $p \neq p'$. It is convenient to enumerate clusters in such a way that $X^{(p)} > X^{(p')}$ for $p > p'$.

Coinciding coordinates in strong clusters require addressing the problems related to discontinuities in the right-hand-side of Eq. (4). We deal with these difficulties using the separability of strong clusters emerging from generic initial conditions and continuity and boundedness of the relaxed cut function. The character of interaction between the relaxed spins in the V_2 model allows one to separate the dynamical contributions stemming from the interactions within the clusters and between the relaxed spins in different clusters, which enables an analysis of the convergence to a system of strong clusters and its stability. To utilize this feature, we introduce *weak clusters*, when groups of relaxed spins can be clearly identified without having all continuous components within a group coinciding. More formally, a system of weak clusters is a partition $\mathscr{V} = \bigcup_p \mathscr{U}^{(p)}(\boldsymbol{\sigma}, \mathbf{X})$ induced by a system of non-intersecting open intervals $\mathscr{I}^{(p)} = \left(X_-^{(p)}, X_+^{(p)}\right) \subset [-1, 1)$, such that $\mathscr{U}^{(p)}(\boldsymbol{\sigma}, \mathbf{X}) = \left\{\alpha \in \mathscr{V} : X_\alpha \in \mathscr{I}^{(p)}\right\}$. We say that a weak cluster is *generic* if all continuous components of the relaxed spins are distinct: $X_\alpha \neq X_\beta$ for all $\alpha, \beta \in \mathscr{V}$ such that $\alpha \neq \beta$.

The next distinctive property of the relaxed cut function $\mathcal{C}_{\mathrm{V}_2}(\boldsymbol{\sigma}, \mathbf{X})$ is that it has no local maxima: its critical points are either saddle points or max-cut. At the same time, the evolution of the V_2 model can terminate in a saddle point. Identifying such situations is the main problem in investigating the convergence of the V_2 model. In this regard, another important feature of the V_2 model is worth emphasizing as it greatly enhances the model computational power. While departing from a slight perturbation of a binary state, the system of relaxed spins ends up in a state with the cut not smaller than that of the initial state. More precisely, if the initial state is $(\boldsymbol{\sigma}(0), \mathbf{X}(0))$, the V_2 model progresses in such a way that $\mathcal{C}(\boldsymbol{\sigma}(t)) \geq \mathcal{C}(\boldsymbol{\sigma}(0))$. Moreover, if $\boldsymbol{\sigma}(0)$ does not yield the maximum cut, $\mathcal{C}(\boldsymbol{\sigma}(0)) < \overline{\mathcal{C}_{\mathscr{G}}}$, the probability of improving cut by a random perturbation of the continuous component is strictly positive (Theorem 6 in [14]). We will call the randomly agitated V_2-machine a system, whose dynamics is described by the V_2 model and whose terminal states are randomly agitated. The beneficial property of random agitations is their ability to break disadvantageous patterns that may emerge due to a peculiar character of the initial condition. For example, the model initialized in state $(\boldsymbol{\sigma}, \mathbf{0})$ with random $\boldsymbol{\sigma}$ will stay in this computationally insignificant state. The same trivial result is obtained for an "agitated" initial state of the form $(\boldsymbol{\sigma}, \kappa\mathbf{1})$, with $|\kappa| < 1$. Random agitations eliminate such pathological effects.

III. GRAPH FORMULATION OF DISCRETE TOMOGRAPHY

The main distinguishing features of the dynamics of the V_2 model are established in relation to the evolution of a system of relaxed spin on graphs toward their ground state. Therefore, the discussion of how the V_2 model solves the discrete tomography problem significantly benefits from reformulating the latter in graph-theoretical terms.

The discrete tomography envelops a great variety of problems related to reconstructing images from limited information [16, 17]. Here, we focus on the canonical problem of recovering a binary image from its projections along a selected family of rays. Usually, this problem is formulated as follows. The binary image is represented by $\hat{s}$, a square $W \times W$ matrix with elements $s_{m,n} \in \{0, 1\}$. The tomographic data is given by two vectors $\mathbf{P}^{(\mathrm{row})}$ (assigning to each row the sum over columns) and $\mathbf{P}^{(\mathrm{col})}$ (assigning to each column the sum over rows) with components $P_m^{(\mathrm{row})} = \sum_n s_{m,n}$ and $P_n^{(\mathrm{col})} = \sum_m s_{m,n}$. Solving a tomography problem means reconstructing the binary image from the tomographic data: finding such distribution of set pixels in the $W \times W$ matrix, whose projections along columns and columns reproduce the tomographic data. It should be noted that, in general, the tomography problem is ill-posed: vectors $\mathbf{P}^{(\mathrm{row})}$ and $\mathbf{P}^{(\mathrm{col})}$ do not necessarily define $\hat{s}$ uniquely [16, 18, 19]. Since the objective of the present paper is to outline the computational capabilities of the V_2-machine, we will understand the tomography problem as finding *any* image $\hat{s}$ that satisfies the tomographic data. From this perspective, it is more important that the tomographic data is *consistent*: there exists at least one image satisfying the data. In what follows, we will assume that this is indeed the case.

Another important condition that we will assume always fulfilled is that the tomographic data is collected along such a system of rays that any two rays may intersect at most at one point. For the example above, this condition is met trivially because each pixel $s_{m,n}$ contributes only to $P_m^{(\mathrm{row})}$ and $P_n^{(\mathrm{col})}$.

To reformulate the tomography problem in graph-theoretical terms, we identify the image pixels with with set $\mathscr{V}$ of nodes and define the binary image as a function on the nodes: $\mathbf{s} : \mathscr{V} \to \{0, 1\}$. Next, the system of rays, $\mathscr{R} = \{\mathscr{R}(r) \subset \mathscr{V}, 1 \leq r \leq R : |\mathscr{V}(r) \cap \mathscr{V}(r')| \leq 1, r \neq r'\}$, is a set of rays $\mathscr{R}(r)$, subsets of nodes such that two different subsets have at most one node in common.

The *tomographic data* is a function $\mathbf{P}_\mathbf{s} : \mathscr{R} \to \mathbb{Z}$ evaluating the pixel-sum along the ray, the total weight of intersection of image $\mathbf{s}$ with the rays, $\mathbf{P}_\mathbf{s} : \mathscr{R}(r) \mapsto P_\mathbf{s}(r) := \sum_{u \in \mathscr{R}(r)} s(u)$. In other words, $P_\mathbf{s}(r)$ is the number of set pixels in the r-th ray traversing image $\mathbf{s}$. Respectively, the *tomography problem* is posed as finding a binary image $\mathbf{s}' : \mathscr{V} \to \{0, 1\}$ with matching tomographic data, $\mathbf{P}_{\mathbf{s}'} = \mathbf{P}_\mathbf{s}$.

In these terms, the canonical formulation corresponds to a set of W^2 nodes arranged as a $W \times W$ square on a 2D integer lattice, with subsets $\mathscr{R}(r)$ representing the rows and the columns of the square. At the same time, the generalized formulation covers other classes of besides reconstructing 2D images given the tomographic data collected along the rows and columns of the image. For example, the generalized reformulation naturally extends to problems with a more complex geometry, such as reconstructing 3D images (see Section V B below).

We seek for a graph, whose maximum cut corresponds reconstructing the binary image satisfying the tomographic data. To this end, we consider a particular ray $\mathscr{R}(r) \in \mathscr{R}$ with $N(r) = |\mathscr{R}(r)|$ nodes and its partitioning given by the spin configuration $\boldsymbol{\sigma}$. Associating spins up with set pixels, we find that the reconstructed single-ray image satisfies the tomographic data if $Q(r;\boldsymbol{\sigma}) := \sum_{\alpha\in\mathscr{R}(r)} \sigma_\alpha + \widetilde{q}_0(r)\sigma_0 = 0$, where $\widetilde{q}_0(r) = N(r) - 2P_{\mathbf{s}}(r)$ is the tomographic data written for the spin representation of the binary image and $\sigma_0 \equiv 1$ is the fixed auxiliary spin. The deviation of the image represented by $\boldsymbol{\sigma}$ from the tomographic data can be quantified by penalty $Q^2(r;\boldsymbol{\sigma})$, which has a direct combinatorial meaning. With ray $\mathscr{R}(r)$, we associate graph $\widetilde{\mathscr{K}}(r)$, a complete graph $\mathscr{K}_{\mathscr{R}(r)}$ on $\mathscr{R}(r)$ with added dominating vertex (auxiliary spin) $\{0\}$ connected to the rest of the nodes by edges with weight $\widetilde{q}_0(r)$. We denote by $\widetilde{\mathscr{R}}(r) = \mathscr{R}(r) \cup \{0\}$ the set of nodes in ray r with added dominating vertex. Configuration $\boldsymbol{\sigma}$ induces cut of $\widetilde{\mathscr{K}}(r)$ that can be written as $\mathcal{C}(\boldsymbol{\sigma})_{\widetilde{\mathscr{K}}(r)} = \overline{\mathcal{C}}_{\widetilde{\mathscr{K}}(r)} - \frac{1}{4}Q^2(r;\boldsymbol{\sigma})$, where $\overline{\mathcal{C}}_{\widetilde{\mathscr{K}}(r)} = (N(r) - P_{\mathbf{s}}(r))^2$ is the maximum cut of $\widetilde{\mathscr{K}}(r)$.

Extending such consideration to all rays, we find that reconstructing binary image from tomographic data is equivalent to finding the maximum cut partition of graph $\mathscr{G}_{\mathbf{s},\mathbf{P}} = \left\{\widetilde{\mathscr{V}}, \mathscr{E}\right\}$, with $\widetilde{\mathscr{V}} = \mathscr{V} \cup \{0\}$, constructed by the union of cliques $\widetilde{\mathscr{K}}(r)$ with $r = 1, \ldots, R$. An example of the graph representing a discrete tomography problem on a 3×3 grid is shown in Fig. 2.

Depending on the context, it may be convenient either to regard a set of individual auxiliary spins associated with different rays, or to contract all the auxiliary spins to a single dominating vertex. During such a contraction, multiple (or parallel) edges may form because a single node may belong to several rays and hence can be connected to multiple auxiliary nodes corresponding to different cliques. Such multiple edges, in turn, are contracted to a single edge with the accumulated weight. For example, let node α belong to rays $\mathscr{R}(r_1)$ and $\mathscr{R}(r_2)$ with the respective $\widetilde{q}_0(r_1)$ and $\widetilde{q}_0(r_2)$, then in $\mathscr{G}_{\mathbf{s},\mathbf{P}}$ with contracted auxiliary spins, node α is connected to the dominating vertex by the edge with weight $\widetilde{q}_0(r_1) + \widetilde{q}_0(r_2)$.

On the other hand, it is convenient to keep the auxiliary spins separated because in this case cliques $\widetilde{\mathscr{K}}(r)$ do not share edges. Consequently, the cut weight produced by partitioning $\mathscr{G}_{\mathbf{s},\mathbf{P}}$ is the sum of cut weights produced by induced partitioning of $\widetilde{\mathscr{K}}_{\mathscr{R}(r)}$, yielding

$$\mathcal{C}_{\mathscr{G}_{\mathbf{s},\mathbf{P}}}(\boldsymbol{\sigma}) = \sum_r \mathcal{C}_{\widetilde{\mathscr{K}}(r)}(\boldsymbol{\sigma}) = \overline{\mathcal{C}}_{\mathscr{G}_{\mathbf{s},\mathbf{P}}} - \frac{1}{4}\sum_r Q^2(r;\boldsymbol{\sigma}), \quad (5)$$

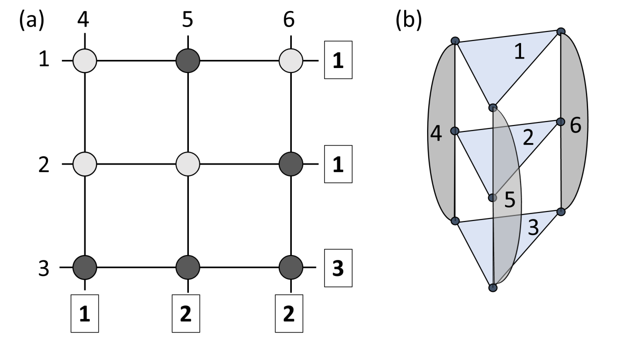

FIG. 2. An example of a graph representation of a discrete tomography problem. (a) A tomography problem set on 3×3 grid. Circles represent pixels, horizontal and vertical lines indicate rays, along which the tomographic data is collected. Plain numbers are an enumeration of rays, and bold framed numbers are tomographic data for the example image (a glider from Conway's Game of Life [20]) with dark shaded circles representing pixels set ON. (b) A graph corresponding to the problem shown in (a). Enumerated shaded regions show cliques representing the corresponding rays and contain auxiliary nodes holding the auxiliary spin.

where $\overline{\mathcal{C}}_{\mathscr{G}_{\mathbf{s},\mathbf{P}}} = \sum_r (N(r) - P_{\mathbf{s}}(r))^2$ is the maximum cut of $\mathscr{G}_{\mathbf{s},\mathbf{P}}$. Here, we observe that the partitioning yielding the maximum cut of $\mathscr{G}_{\mathbf{s},\mathbf{P}}$ simultaneously delivers the maximum cut of all cliques $\widetilde{\mathscr{K}}_{\mathscr{R}(r)}$. This is a graph-theoretical formulation of the consistency of the tomographic data, which plays the key role in the following.

Once a problem is formulated as a max-cut problem for a graph with weighted edges, the equations of motion of the V_2 model can be obtained from graph weighted adjacency matrix using Eq. (4). However, for an analysis of the computational capabilities of the V_2 model, it is constructive to reflect the structure of the graph representing the tomography problem directly in the equations of motion. Therefore, we delay writing the explicit form of these equations till the next section [see Eqs. (15) and (16)], where they will be presented in a much more compact form compared to a straightforward explication of Eq. (4).

IV. CONVERGENCE TO THE SOLUTIONS

A. Terminal states for single-ray problems

The computational capabilities of an Ising machine are determined by the properties of terminal states reachable from a generic initial state. The analysis of the terminal states is significantly simplified by the fact that all critical points of $\mathcal{C}_{\mathrm{V}_2}(\boldsymbol{\sigma}, \mathbf{X})$ belong to critical manifolds enumerable by purely binary states [14]. Therefore, while considering whether a particular terminal state can be reached, the binary component can be considered fixed, and states unstable with respect to small perturbations can be disregarded. In the worst case, the system can be led away from them by random agitations. As we will see, many saddle critical manifolds can be ruled out based on this simple analysis.

However, for some critical manifolds such consideration is not sufficient. The ground for a more refined analysis is the observation that stability of a critical point is determined by the equations of motion of the same structure as of the equations arising in 1D electrostatics with vector charges. To relate these seemingly different situations, we consider the simplest case when the image is covered by a single ray: $\mathscr{R} = \{\mathscr{R}(1)\}$ with $\mathscr{R}(1) = \widetilde{\mathscr{V}}$, or, in other words, graph $\mathscr{G}_{\mathbf{s},\mathbf{P}}$ coincides with clique $\widetilde{\mathscr{K}}_{\mathscr{R}(1)}$. In this case, the tomographic data turn into a single number $P(1)$, the number of activated pixels in the image. With each free node $\alpha \neq 0$, we associate charge $q_\alpha = \sigma_\alpha$, and to the auxiliary node we assign charge $q_0 = \sigma_0 \widetilde{q}(1) = N(1) - 2P(1)$. Using these notations, equations of motion (4) for free spins can be rewritten in a simple form

$$\dot{X}_\alpha = q_\alpha \epsilon\left(X_\alpha\right), \tag{6}$$

where

$$\epsilon\left(X\right) = \sum_{\beta: X_\beta < X} q_\beta - \sum_{\beta: X_\beta > X} q_\beta. \tag{7}$$

Here, the summation runs over all spins, including the auxiliary one.

Next, we notice that if we are interested only in establishing the stability of a given spin configuration $\boldsymbol{\sigma}$, the system of equations (6) can be regarded as defined on the whole real line. It is straightforward to show that spin distribution $\boldsymbol{\sigma}(0)$ is stable with respect to perturbation $\mathbf{X}(0)$, that is the system initially in state $(\boldsymbol{\sigma}(0), \mathbf{X}(0))$ terminates in a state with the same discrete component, if and only if the evolution governed by Eqs. (6) on $\mathbb{R}$ is bound. Consequently, we can investigate the stability of relaxed-spin distributions characterized by the fixed distribution of the discrete component $\boldsymbol{\sigma}$ by using the framework of stability in the Lyapunov sense.

Finally, we observe that the stability of solutions of Eqs. (6) on $\mathbb{R}$ is equivalent to stability of a system of charged particles on 1D (or a system of parallel charged planes) with the same charges, q_i, as defined above for the relaxed spins, and the particle corresponding to the auxiliary spin fixed at the origin. The equivalence follows from the observation that, up to the choice of time scale, the right-hand-side of the equations of motion governing the charged particles has the same form, $q_\alpha \epsilon\left(X_\alpha\right)$, as in Eqs. (6) and (7).

From this perspective, the penalty associated with missing the tomographic constraint can be written as $Q(\boldsymbol{\sigma}) = \sum_{\alpha \in \widetilde{\mathscr{K}}(r)} q_\alpha$ and interpreted as the total ray charge. Respectively, it is constructive to define charges of other collections of relaxed spins, such as strong and weak clusters. When the tomographic constraint is met, the ray charge is zero, $Q(\boldsymbol{\sigma}) = 0$. Since the tomography problem has a solution, the zero value is feasible. Then, it is not difficult to see that $Q(\boldsymbol{\sigma})$ may take only even values. This, simple but important, observation together with the analogy with stability of a system of electric charges immediately yields a key property.

Theorem 1. *For a single ray tomography problem, states with $Q(\boldsymbol{\sigma}) = 0$ (the solutions of the problem) are the only stable states of the V_2 model. Moreover, starting from a generic initial state, V_2 model terminates in a $Q(\boldsymbol{\sigma}) = 0$ state.*

The first part of the theorem is a simple consequence of the fact that at equilibria of the V_2 model, the relaxed and discrete cut functions coincide, and equilibria are either max-cut states or are saddle points. If a terminal state is a max-cut state, then $Q(\boldsymbol{\sigma}) = 0$, and $\boldsymbol{\sigma}$ delivers a solution to the single-ray tomography problem. If the system terminates in a saddle point, then, due to continuity of $\mathcal{C}_{\mathrm{V}_2}(\boldsymbol{\sigma}, \mathbf{X})$, there exist variations of the continuous component of the relaxed spin increasing $\mathcal{C}_{\mathrm{V}_2}(\boldsymbol{\sigma}, \mathbf{X})$. After such displacements, the motion is unbounded because it must lead to a variation of the discrete component of the relaxed spin [14].

The second part of the theorem can be regarded as following from Earnshaw's theorem about stability of a system of electric charges [21] or can be proven directly by respectively adapting the usual argument used to prove Earnshaw's theorem. We start from a useful technical statement

Lemma 1. *Let a weak generic cluster $\mathscr{V}^{(p)}$ occupying interval $\left(X_-^{(p)}, X_+^{(p)}\right)$, such that either $X_-^{(p)} > 0$ or $X_+^{(p)} < 0$ (so that $\mathscr{V}^{(p)}$ does not contain the auxiliary spin) have sufficiently high charge, $\left|Q^{(p)}(\boldsymbol{\sigma})\right| > 1$, where $Q^{(p)}(\boldsymbol{\sigma}) = \sum_{\alpha \in \mathscr{V}^{(p)}} q_\alpha$. Then $\mathscr{V}^{(p)}$ is unstable.*

Proof. According to Eq. (7), the equations of motion governing the relaxed spins inside $\mathscr{V}^{(p)}$ are determined by the field $\epsilon(X_\alpha)$ that can be written as

$$\epsilon\left(X_\alpha\right) = \epsilon^{(\bar{p})} + \epsilon^{(p)}\left(X_\alpha\right), \tag{8}$$

where $\epsilon^{(\bar{p})}$ and $\epsilon^{(p)}\left(X_\alpha\right)$ are the contributions due to the relaxed spins inside and outside of $\mathscr{V}^{(p)}$, respectively. Since $\left|Q^{(p)}\right| \geq 2$, there are two distinct relaxed spins in $\mathscr{V}^{(p)}$ with the charge $q^{(p)}$ of the same sign as $Q^{(p)}$ and extremal values of their X-coordinates among the relaxed spins in $\mathscr{V}^{(p)}$ with the same charge. Let these marginal relaxed spins be β and γ, and $X_- < X_\beta < X_\gamma < X_+$. Then, we have

$$\begin{aligned} \frac{d}{dt}\left(X_\gamma - X_\beta\right) &= q^{(p)}\left[\epsilon^{(p)}\left(X_\gamma\right) - \epsilon^{(p)}\left(X_\beta\right)\right] \\ &= 2q^{(p)}\left[q^{(p)} + \sum_{\alpha: X_\beta < X < X_\gamma} q_\alpha\right] > 0. \end{aligned} \tag{9}$$

The inequality follows from comparing the expression in the square brackets with the total charge of $\mathscr{V}^{(p)}$.

Thus, the separation between β and γ increases leading to either unbound evolution or collision with other clusters. $\square$

Proof of Theorem 1. The whole set of relaxed spins can be regarded as a weak cluster with a charge of even (including zero) magnitude. To apply Lemma 1, we need to incorporate the auxiliary spin with its fixed continuous component.

The argument used for proving Lemma 1 would be invalidated if the auxiliary spin is the only carrier of the excess charge. It is easy to show, however, that this is impossible. Without loss of generality, we can assume that the excess charge is positive, $Q(\boldsymbol{\sigma}) > 0$. Now, suppose the auxiliary spin with $q_0 > 0$ is the only positively charged particle, so that all free relaxed spins have $q_\alpha = -1$. However, in this case, the total charge is $q_0 + \sum_{\alpha\neq 0} q_\alpha = N - 2P - N = -2P \leq 0$, which contradicts the assumption that $Q(\boldsymbol{\sigma}) > 0$.

What remains to be checked is how Eq. (9) changes when one of the marginal relaxed spins is the auxiliary spin. Let $Q(\boldsymbol{\sigma}) > 0$, and the auxiliary spin with $q_0 > 0$ and relaxed spin β with $X_\beta > 0$, are positively charged relaxed spins with the smallest and the largest continuous components, respectively. Then, we have

$$\frac{dX_\beta}{dt} = q_\beta \left[\sum_{\alpha:X<X_\beta} q_\alpha - \sum_{\alpha:X>X_\beta} q_\alpha \right] \geq q_\beta \left[Q(\boldsymbol{\sigma}) - q_\beta \right] > 0, \tag{10}$$

where we have taken into account that $|Q(\boldsymbol{\sigma})|$ is even. Equation (10) is intentionally written in the form valid for both $Q(\boldsymbol{\sigma}) > 0$ and $Q(\boldsymbol{\sigma}) < 0$ cases. It is elementary to adopt the argument to the case when $X_\beta < 0$ and to show that in this case $dX_\beta/dt < 0$.

Thus, generic weak clusters can only converge to a strong cluster if they represent a solution of the tomography problem. $\square$

We conclude the analysis of the single-ray case by noting that except for the cases with all pixels set or unset, single-ray tomography problems have multiple solutions [14]. In this case, the terminal state can be expected to consist of multiple strong clusters. Since Lemma 1 is applicable to each cluster independently, we obtain that the resultant strong clusters must have zero charge. In particular, this implies that if a strong cluster does not contain the auxiliary spin, then, generically, it should comprise only two relaxed spins.

B. Terminal states for multiple rays

A generalization of the approach in the previous section to tomography problems with $R > 1$ rays requires accounting for the fact that the relaxed spins belonging to different rays do not interact with each other. This is achieved by associating with the relaxed spins vector charges defined in the R-dimensional Euclidean space. To this end, in $\mathbb{R}^R$, we introduce basis vectors $\vec{e}(r)$ with coordinates $\vec{e}(r)_j = \delta_{r,j}$. Next, for each relaxed spin α we introduce $\mathscr{D}(\alpha) = \{r : \alpha \in \mathscr{R}(r)\}$, the set of rays containing α. Finally, we define the vector charge for a free relaxed spin $\alpha \neq 0$

$$\vec{q}_\alpha = \sigma_\alpha \sum_{r\in\mathscr{D}(\alpha)} \vec{e}(r). \tag{11}$$

Respectively, for the auxiliary spin, we define

$$\vec{q}_0 = \sum_{r\in\mathscr{D}(\alpha)} \widetilde{q}_0(r)\vec{e}(r). \tag{12}$$

Using these definitions, one can define vector charges of collections of relaxed spins as a sum of vector charges. For example, the total vector charge carried by state $(\boldsymbol{\sigma}, \mathbf{X})$ is $\vec{Q}(\boldsymbol{\sigma}) = \sum_{\alpha\in\widetilde{\mathscr{V}}} \vec{q}_\alpha$, while the total charge of relaxed spins in ray r is natural to define as $\vec{Q}(r;\boldsymbol{\sigma}) = \sum_{\alpha\in\mathscr{R}(r)} \vec{q}_\alpha$. In terms of vector charges, the cut weight of graph $\mathscr{G}_{\mathbf{s},\mathbf{P}}$ [Eq. (5)] takes a simple form

$$\mathcal{C}_{\mathscr{G}_{\mathbf{s},\mathbf{P}}}(\boldsymbol{\sigma}) = \overline{\mathcal{C}}_{\mathscr{G}_{\mathbf{s},\mathbf{P}}} - \frac{1}{4}\vec{Q}(\boldsymbol{\sigma}) \cdot \vec{Q}(\boldsymbol{\sigma}), \tag{13}$$

where $\vec{a} \cdot \vec{b}$ denotes the inner product in Euclidean space. Equation (13) is obtained directly from Eq. (5) by noticing that $Q(r;\boldsymbol{\sigma}) = \vec{e}(r) \cdot \vec{Q}(\boldsymbol{\sigma}) = \vec{e}(r) \cdot \vec{Q}(r;\boldsymbol{\sigma})$.

Respectively, the relaxed cut function [Eq. (2) written for the discrete tomography problem] is written as

$$\mathcal{C}_{\mathrm{V}_2}(\boldsymbol{\sigma}, \mathbf{X}) = \mathcal{C}_{\mathscr{G}_{\mathbf{s},\mathbf{P}}}(\boldsymbol{\sigma}) + \frac{1}{4}\sum_{\alpha,\beta} \vec{q}_\alpha \cdot \vec{q}_\beta |X_\alpha - X_\beta|. \tag{14}$$

Finally, the equations of motion governing the V_2 model are a straightforward generalization of Eqs. (6) and (7):

$$\dot{X}_\alpha = \vec{q}_\alpha \cdot \vec{\epsilon}(X_\alpha)\,, \tag{15}$$

where

$$\vec{\epsilon}(X) = \sum_{\beta:X_\beta<X} \vec{q}_\beta - \sum_{\beta:X_\beta>X} \vec{q}_\beta. \tag{16}$$

A substantial technical deviation from the single-ray case is expressed by the fact that in the vector case, the formal equilibrium corresponds not only to the case of vanishing vector field but also to the case of orthogonal $\vec{q}_\alpha$ and $\vec{\epsilon}(X_\alpha)$. Dynamically, the orthogonality expresses the exact cancellation of the contributions originating from different rays containing the relaxed spin. Such cancellation invalidates an important property self-evident in the single-ray case that in equilibrium there are no isolated relaxed spins.

The accidental orthogonality can be eliminated by ensuring that projections $\vec{e}(r) \cdot \vec{Q}(r)$ for different r are linearly independent over $\{-1, 1\}$. In our simulations, such linear independence was forced by redefining the

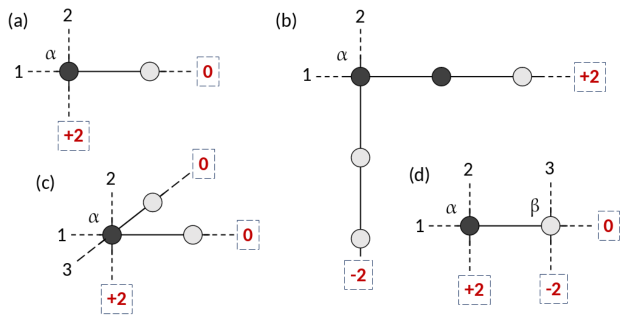

FIG. 3. Examples of configurations violating the tomographic data that are stable equilibria for the relaxed spin α. Dashed-framed numbers show deviations of the respective components of the vector charge from zero due to mismatched tomographic data. Circles depict states of the discrete components that need to be corrected. (a) Spin α is in neutral equilibrium. (b) Spin α is asymptotically stable (there is a returning field acting on α). Five spins need to be corrected to match the tomographic data. (c) A configuration similar to (a) but at the intersection of three rays. Spin α is asymptotically stable state, while only three spins need to be corrected. (d) A realization of the configuration from (a) for a tomography problem on a 2D grid leads to violating the tomographic data in two rays. Both α and β are in neutral equilibrium.

last term in Eq. (14) as $\frac{1}{4}\vec{Q}(\boldsymbol{\sigma})\cdot\Lambda\cdot\vec{Q}(\boldsymbol{\sigma})$, where $\Lambda = \mathrm{diag}(\lambda(1),\ldots,\lambda(R))$ is a diagonal matrix with positive linearly independent $\lambda(r) > 0$. Alternatively, this procedure can be understood as redefining $\vec{e}(r) \to \sqrt{\lambda(r)}\vec{e}(r)$. In computations, it was sufficient to generate the diagonal elements as $\lambda_r = \sqrt{p(r)}/\lfloor\sqrt{p(r)}\rfloor$, where $\{p(r)\}$ is a set of random distinct prime numbers.

Satisfying the tomographic data for the r-th ray results in $\vec{e}(r)\cdot\vec{Q}(r) = 0$. This provides a way for translating the results obtained in the previous section to the multi-ray case and identifying the main barriers for converging to a reconstructed image starting from generic initial conditions and the dynamical features of the V_2 model that allow overcoming these barriers.

The conclusion that the V_2 states $(\boldsymbol{\sigma},\mathbf{X})$ with $\boldsymbol{\sigma}$ satisfying the tomographic data are the only stable terminal states remains valid for the problems with multiple rays as well. On the other hand, the argument used for proving the second part of Theorem 1 is invalid. This reflects the fact that for multiple-ray tomography problems, Earnshaw's-like statements about instability of *individual* charges are no longer true for vector charges. As demonstrated by Fig. 3, there do exist arrangements of vector charges stable with respect to any single-particle disturbances. Respectively, to match the tomographic data in the presence of these patterns requires inversion of *multiple* spins. Consequently, local search-based approaches to solve the spin formulation of the discrete tomography problem are prone to terminate in states with such defects as manifested in numerical simulations shown in Section V A.

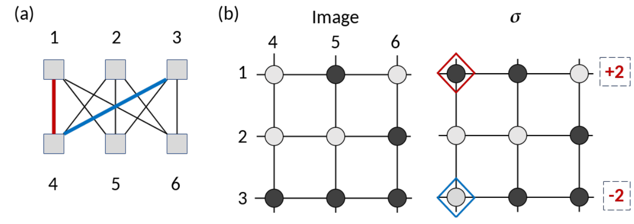

FIG. 4. (a) An example of the ray graph $\mathscr{G}_{\mathscr{R}}$ for the discrete tomography problem on the 3×3 grid. The highlighted edges show edge-induced subgraph $\mathscr{T}_{\mathscr{R}}(\boldsymbol{\sigma}'|\boldsymbol{\sigma}) \subseteq \mathscr{G}_{\mathscr{R}}$. (b) An example of a 3×3 image and an incorrect spin configuration stable with respect to single-spin perturbations. Correction of the selected spins is represented by $\mathscr{T}_{\mathscr{R}}(\boldsymbol{\sigma}|\boldsymbol{\sigma})$ shown in (a).

In turn, inherently, the V_2 model's trajectories are not constrained to Hamming-limited variations of the discrete component of the relaxed spin [14]. As a result, the V_2-machine can either avoid defects or escape them after the random agitation of a terminal state breaking the tomographic data.

To discuss this property of the V_2 model, we limit ourselves to the tomography problem on a 2D rectangular grid, where the typical defects in terminal states stable with respect to single-particle perturbations have the form shown in Fig. 3. It is not difficult to see that such patterns lead to mismatching the tomographic data in, at least, two rays.

Lemma 2. *For a tomographic problem on a rectangular grid, if the spin configuration $\boldsymbol{\sigma}$ does not match the tomographic data, $\vec{Q}(\boldsymbol{\sigma})\cdot\vec{Q}(\boldsymbol{\sigma}) > 0$, then there are at least two rays with unsatisfied tomographic data.*

Proof. We represent the system of rays by graph $\mathscr{G}_{\mathscr{R}}$, whose nodes correspond to rays, and two nodes are connected if the respective rays intersect. For example, the graph representing the tomography problem for a $W\times W'$ image is a complete bipartite graph $\mathscr{K}_{W,W'}$ with partitions, $\mathscr{A}$ and $\mathscr{B}$, corresponding to rows and columns, respectively (see Fig. 4). We notice that the edges in this graph correspond to the image pixels. Indeed, two rays intersect only at a single pixel, and each pixel lies at the intersection of at most two rays, thus there is a bijection between pixels and the edge set of $\mathscr{G}_{\mathscr{R}}$. Having such correspondence established, we assign to the edges weights ± 1 according to the value of the respective spins. In other words, we regard the spin configuration $\boldsymbol{\sigma}$ as defining a function $\boldsymbol{\sigma} : \mathscr{E}(\mathscr{G}_{\mathscr{R}}) \to \{-1, 1\}$.

Let the tomographic constraint be broken in ray r_1 by having one too many set pixels, so that $Q(r_1;\boldsymbol{\sigma}) = \vec{e}(r_1)\cdot\vec{Q}(r_1;\boldsymbol{\sigma}) = 2$. Let the error be correctable by inverting relaxed spin $\alpha \in \mathscr{R}(r_1)$ with $\sigma_\alpha = 1$. This relaxed spin lies at the intersection of rays $\mathscr{R}(r_1)$ and $\mathscr{R}(r_2)$ and, by assumption, $Q(r_2;\boldsymbol{\sigma}) = 0$. Thus, there another relaxed spin $\beta \in \mathscr{R}(r_2)\cap\mathscr{R}(r_3)$ with $\sigma_\beta = -1$ must be inverted together with σ_α to maintain zero error in $\mathscr{R}(r_2)$. Consequently, there exists relaxed spin $\gamma \in$

$\mathscr{R}(r_3) \cap \mathscr{R}(r_4)$ with $\sigma_\gamma = 1$ that must be inverted to maintain zero error in $\mathscr{R}(r_3)$, and so on.

This observation shows that a transformation from the configuration $\boldsymbol{\sigma}$ to a correct configurations $\boldsymbol{\sigma}'$ selects a subset of edges in $\mathscr{G}_{\mathscr{R}}$ corresponding to the set of spins that need to be inverted. This subset defines an edge-induced subgraph $\mathscr{T}_{\mathscr{R}}(\boldsymbol{\sigma}) \subseteq \mathscr{G}_{\mathscr{R}}$. For each node $\mathscr{R}(r) \in \mathscr{T}_{\mathscr{R}}(\boldsymbol{\sigma}'|\boldsymbol{\sigma})$, the total variation of $Q(r;\boldsymbol{\sigma})$ is equal to the doubled total weight of the edges incident to $\mathscr{R}(r)$. Since, as was discussed above, the tomographic data may not uniquely define the image, there may be multiple graphs $\mathscr{T}_{\mathscr{R}}(\boldsymbol{\sigma})$ describing the respective correcting transformations. Since, by the main assumption, the tomography problem has a solution, there exists at least one such graph, which is sufficient for our purposes.

First, we notice that if the degree of $\mathscr{R}(r)$ for some r is odd, then $|Q(r;\boldsymbol{\sigma})| > 0$. But there must be an even number of nodes with odd degree in a graph, implying the lemma.

It remains to consider the case when $\mathscr{T}_{\mathscr{R}}(\boldsymbol{\sigma})$ is Eulerian. In this case, the existence of more than one ray mismatching the tomographic data follows from the fact that there are no odd cycles in $\mathscr{T}_{\mathscr{R}}(\boldsymbol{\sigma})$. To employ this property, we introduce an orientation of edges in $\mathscr{T}_{\mathscr{R}}(\boldsymbol{\sigma})$. Without loss of generality, we assume that the tomographic constraint is broken in ray $\mathscr{R}(r_1) \in \mathscr{A}$ and $Q(r_1;\boldsymbol{\sigma}) > 0$. Then, we define edges with positive and negative weights as oriented from $\mathscr{A}$ to $\mathscr{B}$ and from $\mathscr{B}$ to $\mathscr{A}$, respectively. Then, due to the absence of odd cycles, any directed cycle defines such a set of relaxed spins whose inversion does not change the ray charges and, therefore, cannot correct the mismatched tomographic data. Such directed cycles can be removed without loosing the property of $\mathscr{T}_{\mathscr{R}}(\boldsymbol{\sigma})$ to describe a correcting transformation.

Next, we construct a directed trail starting from $\mathscr{R}(r)$ following the orientation of the incident edges. Any return to $\mathscr{R}(r)$ defines a directed cycle, which can be removed from $\mathscr{T}_{\mathscr{R}}(\boldsymbol{\sigma})$ without exhausting the edges outgoing from $\mathscr{R}(r)$. Without directed cycles, a directed trail starting from $\mathscr{R}(r)$ must terminate at another ray $\mathscr{R}(r')$ with an imbalance between incoming and outgoing edges and, hence, $|Q(r';\boldsymbol{\sigma})| > 0$.

It is worth noticing that the case of Eulerian $\mathscr{T}_{\mathscr{R}}(\boldsymbol{\sigma})$ corresponds to strong violations of the tomographic constraint, $|Q(r;\boldsymbol{\sigma})| \geq 4$ and $|Q(r';\boldsymbol{\sigma})| \geq 4$. Adapting the argument used in the proof of Lemma 1, it can be shown that the respective configurations $\boldsymbol{\sigma}$ cannot emerge from a weak cluster. □

In view of Lemma 2, we limit ourselves to considering the simplest case of a defect shown in Fig. 3(d) and showing that random agitations can lead to rearranging of both spins α and β eliminating errors in both rays. It is not difficult to see that it is sufficient that after the random agitation, relaxed spins α and β both have, say, positive continuous components maximal compared to other relaxed spins in rays containing α and β. Indeed, in this case, the equations of motion of these relaxed spins take the form

$$\begin{aligned}
\frac{dX_\alpha}{dt} &= \vec{q}_\alpha \cdot \left(\vec{Q}(\boldsymbol{\sigma}) - \vec{q}_\alpha - \vec{q}_\beta\right) + \vec{q}_\alpha \cdot \vec{q}_\beta \operatorname{sgn}\left(X_\alpha - X_\beta\right), \\
\frac{dX_\beta}{dt} &= \vec{q}_\beta \cdot \left(\vec{Q}(\boldsymbol{\sigma}) - \vec{q}_\alpha - \vec{q}_\beta\right) + \vec{q}_\alpha \cdot \vec{q}_\beta \operatorname{sgn}\left(X_\beta - X_\alpha\right).
\end{aligned} \tag{17}$$

Taking into account that $\vec{Q}(\boldsymbol{\sigma}) = 2\left(\vec{q}_\alpha + \vec{q}_\beta\right)$, we find

$$\frac{d}{dt}\left(X_\alpha + X_\beta\right) = \left(\vec{q}_\alpha + \vec{q}_\beta\right)^2 > 0. \tag{18}$$

Considering other particles in the rays containing α and β with the X-coordinates taking the maximal (after excluding α and β) value, one can easily check that they are non-increasing. Thus, α and β remain outsiders with increasing X-coordinates, hence their motion is unbound. For the V_2 model evolution, this implies that both spins α and β will invert, which eliminates the error in the recovered image.

This observation allows us to estimate how the number of random agitations required to converge to a solution scales with the image size. To have the scenario reviewed above, after random agitation, the continuous components of the relaxed spins carrying faulty states should be at the extreme positions compared to the remaining relaxed spins in their respective rays. Consequently, the probability to have such an outcome after random agitation is $P_{\text{improve}} \sim \frac{1}{W^2} = \frac{1}{N}$, implying the polynomial scaling of finding a solution with increasing the image size. Numerical simulations presented in the next section show that this estimate is overly pessimistic indicating that correcting the defects may follow other scenarios besides considered above.

V. NUMERICAL DEMONSTRATIONS

The theoretical framework developed above shows the origin of the V_2 model capability to solve discrete tomography problem. At the same time, solving practical tomography problems requires much more than recovering *an* image from tomographic data. For example, one may impose additional constraints on the recovered image (see, e.g. [22]) or accounting for possible errors in the tomographic data itself [23], and so forth. In the present paper, we limit ourselves to demonstrations of the basic capabilities of the V_2 model on two simplified examples: reconstructing 2D and 3D images. The evolution of the V_2 model was simulated using the Euler approximation supplied with the rule of updating the binary component of the relaxed spin as illustrated in Fig. 1. While the Euler approximation is not trouble-free and may lead to emergence of various artifacts [24], it is sufficient for our purposes in the present paper. In all the examples, the V_2-machine ran unsupervised without any additional processing besides applying random agitations.

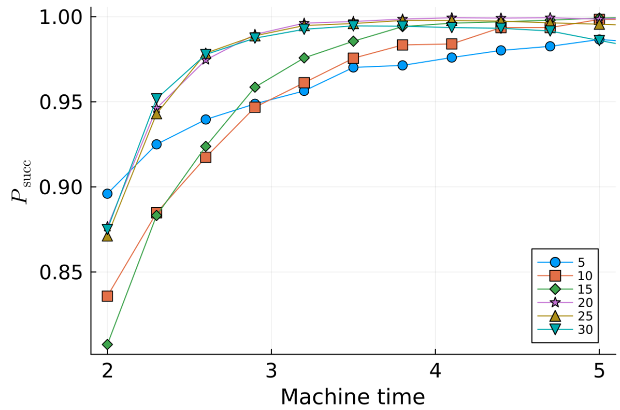

FIG. 5. Probability of successful reconstruction of images $P_{\rm succ}$ from the tomographic data for square random binary images with the pixel activation probability $p = 0.5$ as a function of the duration of the free evolution stage between random agitations. The number of agitations is limited to four. The results are shown for random images with the selected image width: $5, 10, 15, 20, 25, 30$.

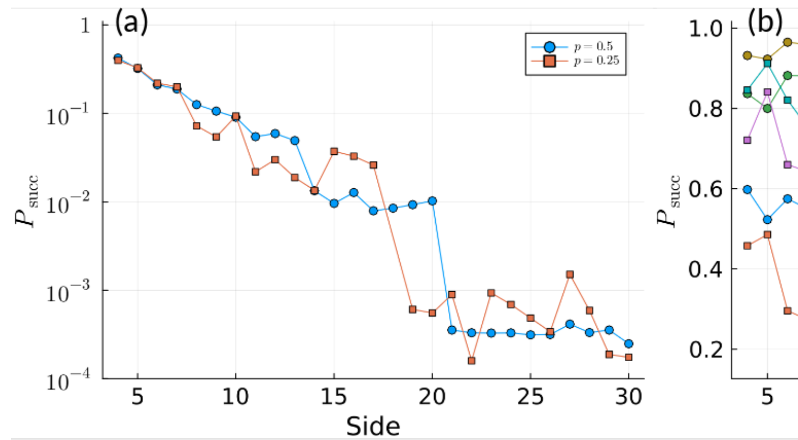

FIG. 6. Probability of successful reconstruction of images $P_{\rm succ}$ from the tomographic data for square random binary image. Circles and squares denote data for random images with the pixel activation probability $p = 0.5$ and $p = 0.25$, respectively. The data points are obtained by averaging over five random images. (a) Using 1-opt local search. Because of a strong variation of the success probability, the plots should be regarded as general trend indicators. (b) Using the V_2 model with $a = 0$, $a = 1$, and $a = 2$ random agitations following the free evolution stage.

A. Tomography of 2D binary images

As the first demonstration, we studied the probability $P_{\rm succ}$ of successful reconstruction of an image from its tomographic data on a set of random square binary images with the sides varying from 4 to 85. For each size, two types of images were generated with the probability for each pixel to be activated $p = 0.5$ and $p = 0.25$. For both image types, $P_{\rm succ}$ was obtained by averaging over the set of five images.

The convergence probability of the V_2 model simulated using the Euler approximation is essentially determined by only two hyperparameters: the duration of the free evolution stage between random agitations T and the magnitude of the time-step, $\Delta t = \frac{T}{M}$, where M is the number of time-steps within the free evolution stage. To demonstrate the dependence on T, Fig. 5 shows $P_{\rm succ}(T)$ with time measured in units defined by the model equations of motion (15). The success probability was obtained by sampling terminal states starting from 10^3 random initial conditions and with the number of random agitations fixed to four. The dependence $P_{\rm succ}(T)$ demonstrates the emergence of several universal dynamical features. They were utilized in the simulations presented below by setting $T = 5$ across all simulations. It should be noted that as the number of time steps covering the free evolution stage was kept constant, $M = 600$, the errors related to the duration of the time step become apparent for larger images at $T \gtrsim 5.0$. The adverse impact of too coarse time-steps Δt is exacerbated in large images as will be discussed below.

The wall-time required by a hardware or a software implementation to process one agitation depends on organization of computation. The presented simulations utilized the sparsity of the graph $\mathscr{G}_{\mathbf{s},\mathbf{P}}$ and were performed in a single-thread. Thus, the time complexity of the V_2 simulations can be estimated as $\mathcal{O}(MW^3)$, where W is the width of the field, per one agitation stage. An efficient parallelization of computations can be expected to reduce the wall-time scaling to $\mathcal{O}(MW)$ per agitation stage.

Figure 6 compares two approaches to recovering an image from its tomographic data using the spin-based formulation of the tomography problem: using 1-opt local search (greedy search) that ensures that in the terminal state the cut weight cannot be improved by inverting any single spin and the V_2 model. The success probability was sampled for the V_2 model by restarting it from 200 random initial conditions. Since the successful recovery by the local search turns with increasing the image size into a rare event, for the local search, $P_{\rm succ}$ was sampled by restarting it from 10^6 random initial states. While studying the performance of local search on a spin representation of the discrete tomography problem is not our objective, we would like to notice a curious behavior of $P_{\rm succ}(W)$: it drops significantly when the most probable total mismatch with the tomographic data increases by two, or the number of defects like shown in Fig. 3(d) increases by one, but changes only moderately between these transitions. In turn, the variation of the probability to find a solution by the V_2 changes only moderately across the same range of image sizes and demonstrates the significant increase of $P_{\rm succ}$ with the number of random agitations.

Figure 6(a) demonstrates the inherent complexity of the discrete tomography problem. Its spin representation

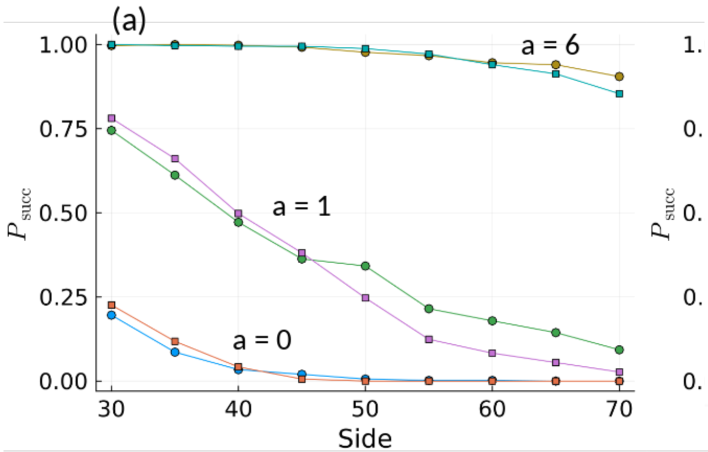

FIG. 7. Probability of successful reconstruction of images $P_{\rm succ}$ for larger images. (a) Circles and squares denote data for random images with the pixel activation probability $p = 0.5$ and $p = 0.25$, respectively. (b) Images with $p = 0.25$ clearly demonstrate the adverse effect of insufficiently fine time-steps in the Euler approximation. Circles and squares correspond to simulations using $M = 700$ and $M = 900$ per free evolution stage (of fixed duration $T = 5$), respectively. While the regime with 9 random agitations is capable to demonstrate high probability, once Δt is not fine enough, its solving capabilities are completely destroyed by insufficiently precise simulations.

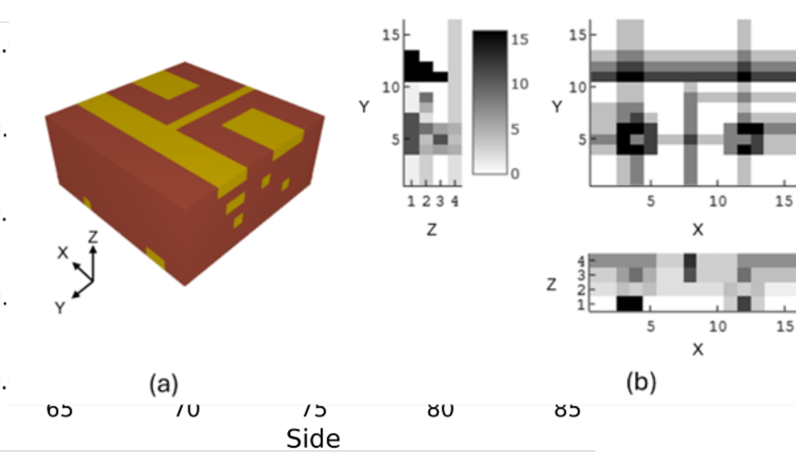

FIG. 8. (a) External view of a 4-layered PCB; (b) model tomographic data collected along the X, Y and Z axes; (c) reconstructed PCB layers

alone does not determine whether an image can be successfully recovered from its tomographic data by a particular implementation of the Ising machine. The ability of the V_2 model to solve the tomography problem is a result of its specific dynamical properties, as discussed in the previous sections.

To demonstrate the scaling properties of the V_2 solving the discrete tomography problem, Fig. 7 shows the variation $P_{\rm succ}(W)$ with the number of random agitations. The image width determines the number of nodes in a single ray and, consequently, the number of nodes in cliques representing rays in graph $\mathscr{G}_{\mathbf{s},\mathbf{P}}$ and the weight of the edges connecting the nodes to the auxiliary spin. As a result, the impact on the rate of change of the continuous component of an individual relaxed spin increases with the image width. As demonstrated in Fig. 7, this effect can be compensated by respective scaling of the number of time-steps M with the image size.

B. Tomography of 3D binary images

As a demonstration of advanced capabilities of the V_2 dynamics, we consider reconstruction of a multi-layered printed circuit board. This problem has a straightforward connection with the general discrete tomography problem, as any PCB (printed circuit board) without integrated circuit chips and other discrete elements can be modelled a three-dimensional binary matrix composed of elements $\{0, 1\}$, where '1' represents the presence of conducting material. We restrict the scale of this demonstration so that the number of required spins is in the order of 10^3 and hence, consider the reconstruction of a small 4-layered PCB, such that the maximum size of the projection image is 16×16 pixels. Moreover, we assume that the ray projection data is available along only three dimensions of the object.

Figure 8a shows the section of an artificial 4-layered PCB we consider for reverse engineering with the conducting tracks depicted in yellow and the non-conducting base in brown. Figure 8b shows the mock tomographic data of the PCB in the YZ, XZ, and XY planes collected by evaluating the pixel-sums along the X, Y, and Z axes, respectively. Figure 8c shows reconstructed PCB layers.

VI. CONCLUSION

We investigated the capability of a relaxation-based dynamical Ising machine driven by the V_2 dynamical model to solve the discrete tomography problem about recovering a binary image from its tomographic data, the pixel sum along a discrete set of rays.

We show that in the simplest single-ray case, the convergence of the V_2 model is governed by the same principles as stability of a system of electrostatically interacting charged particles in 1D. Based on this observation, we show that the V_2 model terminates in a solution to the tomography problem starting from generic initial conditions.

We generalized this approach to the multiple-ray case by identifying a vector charge with the relaxed spins. We identified the main bottleneck terminal states that do not solve the tomography problem, but the V_2 may converge to. We show that random agitations can move the system away from these states, and, as the numerical simulations show, only a few such agitations are needed to ensure the convergence with a high probability ($P_{\text{succ}} \approx 1$). Two key properties of the V_2 model are responsible for this. First, random agitations do not worsen the spin configuration in the terminal state of the system of the relaxed spins. Second, spin transitions occurring during the evolution of the V_2 model are not Hoffman-limited. This should be contrasted to local search-based treatments of the spin reformulation of the discrete tomography problem, for which such transitions present a strong barrier and lead to the quick deterioration of performance with the problem size manifested in numerical simulations.

Our consideration demonstrates that specific dynamical models can be approached as unconventional algorithms producing exact solutions to highly nontrivial data processing tasks. This advances the conventional approach to dynamical Ising machines, regarding them as heuristic devices targeting approximate solutions to optimization problems.

ACKNOWLEDGMENTS

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